



Software Requirements

Specification

for

DAIICT Green

Campus

Management System

Version : 1.2

Group : F4

Vinas Moradiya - 202403060

Preet Hirapara - 202403013

Instructor : Minal Bhise

Course : MC212 - Database Management System

1.Introduction

1.1 Purpose

The purpose of this Software Requirements Specification document is to provide a comprehensive and detailed outline of the software requirements for the DAIICT green campus management System. This document aims to establish a common understanding of the system's objectives, scope, and functionality. Additionally, it acts as a foundation for the design, development, testing, and successful implementation of the DAIICT green campus Management System. The primary aim of the document is to detail out describe the whole system and clearly list all its functionalities.

- The purpose of this project is to design and develop a **Green Campus Management Database System** for DAIICT that maintains comprehensive records of the institute's plantations, tree types, gardener's details and duties, gardening activities, maintenance equipment, which type of fertilizers used, and also we sell our fertilizer which made by our institute so we can manage its all details regarding selling, marketing, raw materials etc.
- Currently, data related to trees, shrubs, herbs, fertilizer and other greenery on the campus is managed manually by staff, this often leads to inefficiencies in monitoring plant health and age, tracking fertilizer usage, and scheduling maintenance activities.
- The system will also support the commercial aspect of DAIICT's green campus by maintaining records of flowers, fertilizers, and other by-products produced on campus, thereby enabling their efficient sale and contributing to revenue generation as well as promoting sustainability.

- **Promote Green & Digital Identity of DAIICT**
 - Strengthen DAIICT's reputation as a technologically advanced and eco-conscious institute.
 - Showcase the campus's biodiversity efforts through digital transparency.
- **Improve Decision-Making for Administration**
 - Generate reports on fertilizer consumption, equipment usage, costs, and plant distribution.
 - Help administrators allocate budget, manpower, and resources more effectively.

The database system will serve as a centralized platform to:

- Maintain accurate records of all plant species (trees, shrubs, herbs, climbers).
- Record the location, growth, and health condition of each plant.
- Track gardening staff responsibilities and schedules.
- Manage equipment inventory required for maintenance.
- Record and optimize fertilizer usage for sustainable growth.

1.2 Intended Audience and Reading Suggestions

This document is intended for the following groups:

- Database Developers / Project Team
 - To understand requirements, design schema, and implement queries.
- Campus Management Staff
 - To use the system for entering, updating, and retrieving plant and maintenance records.
- DAIICT Administration
 - To monitor sustainability efforts, track expenses, and ensure effective resource utilization.
- Researchers and Students
 - To analyze biodiversity, plant growth data, and evaluate campus greening projects.

Reading Suggestions:

- Developers and testers should focus on the technical requirements and system functionality.
- Gardening staff should focus on the features that support daily operations and record-keeping.
- Administration should focus on how the system improves management, monitoring, and decision-making.
- Researchers and students should focus on the data access and reporting features that can support study and analysis.

1.3 Product Scope

- **Plant and Tree Management**

- Store details of every plant: ID, species (botanical/common name), type (tree/shrub/herb), location, date of planting, and health condition.
- Link plants to **QR codes** for real-time information retrieval.
- Record plant growth history and periodic inspections.

- **Fertilizer & Resource Management**

- Maintain fertilizer inventory with purchase details, quantity, and stock availability.
- Track fertilizer application: date, quantity used, plant/area treated.
- Optimize usage to minimize waste and ensure sustainable practices.

- **Equipment Management**

- Maintain an inventory of gardening tools and machinery (sprinklers, pipes, lawnmowers, cutters).
- Record usage history, maintenance schedules, and replacement needs.

- **Gardening Staff Management**

- Maintain records of staff members assigned to different campus zones.
- Track daily tasks, responsibilities, and work shifts.
- Assign plant-care responsibilities and log maintenance activities.

1.4 Description

DAIICT has a large and eco-friendly green campus with thousands of trees, shrubs, herbs, and other plants that require continuous care and maintenance. Currently, much of the record-keeping related to plantation, fertilizers, equipment, and gardening staff is managed manually, which often leads to inefficiency, data loss, and lack of structured reports.

A digital Green Campus Management Database is required to systematically manage and track all activities related to campus greenery. The database will serve as a single point of information for plants, staff, fertilizers, and equipment. It will also support commercial activities such as selling flowers and fertilizers produced on the campus.

1. Plant Management

- Each plant must have a **unique ID**.
- Information to be stored: botanical name, common name, type (tree, shrub, herb, flower plant, etc.), date of planting, location on campus, health status, and growth details.
- Each plant may be linked with a QR code for quick identification and digital access.
- Plants may require multiple fertilizers and periodic maintenance.

2. Fertilizer Management

- Fertilizers need to be tracked with details such as fertilizer ID, type (organic/chemical), quantity in stock, purchase date, expiry date, and cost.
- Fertilizer application records must be maintained: which plant it was applied to, in what quantity, and on what date.

- Fertilizers produced on campus should also be recorded separately for both internal use and sale to external buyers.

3. Equipment Management

- The system must record gardening equipment such as lawnmowers, sprinklers, pipes, and cutting tools.
- Each equipment item will have an equipment ID, name, purchase date, condition, last maintenance date, and assigned staff.
- Usage logs and maintenance history must be tracked.

4. Gardening Staff Management

- Staff members should be recorded with staff ID, name, contact details, and work shift.
- Each staff member may be responsible for maintaining specific plants, applying fertilizers, or handling certain equipment.
- Staff performance and responsibilities must be logged.

5. Sales and Commercial Records

- Flowers, fertilizers, and other by-products produced on campus may be sold.
- The system should record sales transactions, including product type, quantity, buyer details, date of sale, and total cost.
- This ensures transparency and supports revenue tracking.

- Branches collected during pruning or tree maintenance can be sold as raw wood, especially to local artisans, raw material for small industries and biomass energy.
- Small branches, twigs, and leaves can be converted into biomass pellets or **compost**, providing an additional eco-friendly revenue channel.



- By capturing every product of tree care - flowers, leaves, branches, wood, the system ensures that nothing goes to waste and every resource contributes to income or sustainability.

6. Reporting and Analytics

- The system should generate reports on:
 - Number and type of plants on campus.
 - Fertilizer usage patterns and stock status.
 - Equipment maintenance cycles.
 - Staff work allocation and responsibilities.
 - Sales of flowers and fertilizers.
- Reports will help administrators make **data-driven decisions** for campus sustainability.

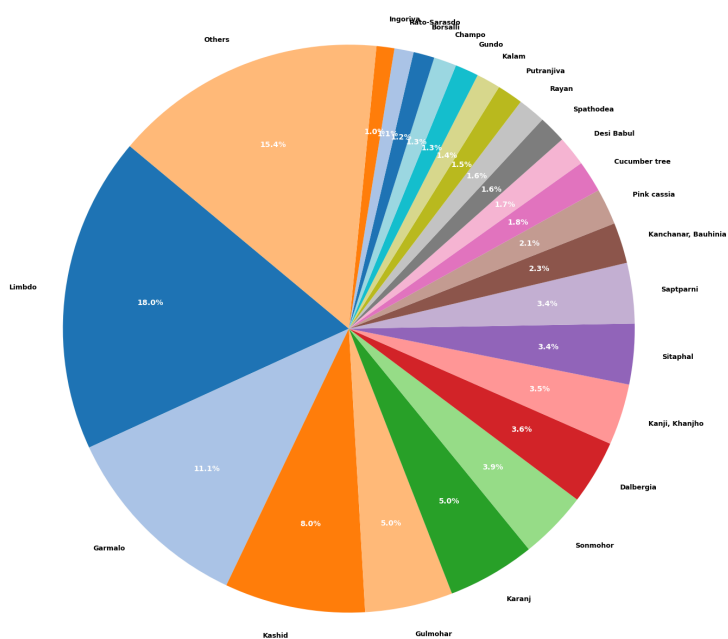
Summary of Plant Species:

Total number of plant available in the campus	5364
Total number of species available in campus	107
Maximum number of plants of same specie available in campus	909

*As per a report of April,2023 prepared by College of Forestry Navsari Agricultural University (NAU), Navsari and DAIICT, Gandhinagar.

As you can see, DAIICT has many trees and plants with many species so we have to manage all trees in an optimal way so we have to manage it by database management system.

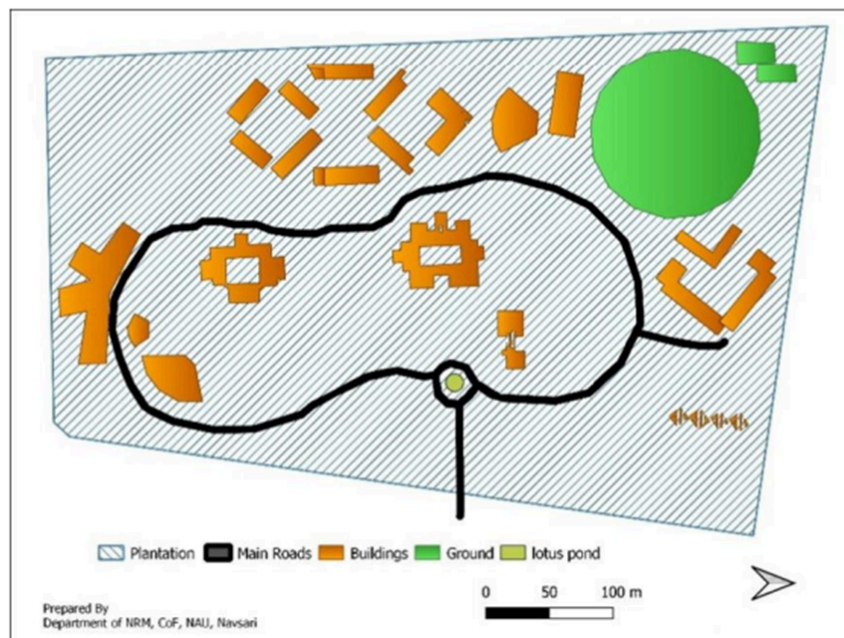
Tree Species Distribution in DA-IICT Campus (Pie Chart with Inside Labels)



General Information about DAIICT :

Latitude	23.188538
Longitude	72.628715
Altitude	81 m
Annual Rainfall	805 mm
Maximum Temperature	36 to 42 °C
Minimum Temperature	10 to 27 °C
Agro climatic situation	North-Central-East Gujarat
Location	Gandhinagar
Soil type	Alluvial clay loam to sandy clay
Topography	Plain

The plain topography supports plantations of shade trees, flowering trees and sustainability-focused trees. These trees help in carbon sequestration, biodiversity support, and revenue generation through flowers, fruits, compost, and wood byproducts.



In the DAIICT campus map, you can see that most of the area is covered with plantations, so it needs to be managed in an optimal way.

QR code generation of each trees and plants:

We need to create a unique QR code that can redirect to a unique tree ID, and based on that tree ID, it will open an html page containing all the information about that tree.

For Example,

Tree ID: A1342



By scanning this QR code, that fetch particular tree ID and give information about that particular tree.



DA-IICT – Trees details

Tree ID.	: A555
Botanical name	: <i>Cassia fistula</i> Linn.
English name	: The purging fistula, Indian laburnum
Common / Local Name	: Garmalo
Family	: Caesalpiniaceae
Uses	: Timber used as posts, Shafts, Carts, Furniture, Boat making, Fuelwood, Charcoal, Medicinal
Photo URL	: Click here for Photo.
Wikipedia Link	: Click here for more details.



DA-IICT – Trees details

Tree ID.	: A1001
Botanical name	: <i>Bismarckia nobilis</i> Hildebr. & H. Wendl.
English name	: Bismarckia
Common / Local Name	: Bismarckia
Family	: Arecaceae
Uses	: Landscaping, ornamental
Photo URL	: Click here for Photo.
Wikipedia Link	: Click here for more details.

2.Requirements Collection / Fact Finding Phase

2.1.1 Description Of Background Readings:

Green Design and Environmental Conservation at DA-IICT: Amitabh Teatia's Landscape Architecture:

The architecture of DA-IICT is functional, but what surrounds it is a fascinating garden. The entire design was oriented to "Preserve the Environment". The different blocks are located such that there is minimal damage to trees, and alongside that the majority of the 2566 trees chosen are indigenous varieties. The lawn area measures roughly 50,000 square meters with 3900 square meters of wild St Augustine grass and 3400 Bamboos were planted to mask the concrete residential and educational buildings.

Next to the library stands the campus herb garden with different varieties of rare medicinal plants which can be used by both the students and staff alike. The entire landscape was planned and developed in a manner so that no excess rainwater is lost and all irrigation is carried out with recycled water from the small on site lake, which along with the fountain have been strategically located so that the wind blows the cooler air across the campus. Water is of particular importance because although the region is not necessarily short on it due to the northerly regions supplying the rivers the area itself has minimal rainfall even during the monsoon season, so being as self sufficient as possible is an advantage. This factor combined with the shade that the existing and newly planted trees give the area are a considerable advantage for both the 30+ species of birds which are observed within the confines of the campus and for students and visitors to relax during the summer heat when temperatures can reach in excess of 45°C (113°F).

With the campus located in India's hot dry desert region to the west the garden was designed in a way that the atmosphere remains cool and calm. Trees were planted so that the noise from the exterior roads was reduced

so students could study in a natural peaceful environment. Benches are scattered throughout the campus to promote external studying and recreation as well as being located in the parking areas for those waiting for transportation or for smokers away from the interior sections of the site.

Prior to the university the area was mostly a forested area and once our design team took control of the landscape side of the project we halted the further reduction of trees from the landscape. This is why many of the trees can be found to be situated in the middle of pathways or in other unusual positions throughout the campus. This not only serves as an environmental advantage by maintaining already mature trees but it also adds numerous unique and quirky features to the campus. Along with the trees within the bath, most pathways were designed to have heavy plantations on both sides to help transport students from the educational environment if only for a short while but aiding in the reduction of stress levels.

Green Campus Programme By Green-Campus_Ireland:

The Litter and Waste theme is the compulsory starting point for all campuses participating in the Green-Campus Programme. It focuses on assessing the impact of waste generation and littering within the institution and exploring practical solutions for reduction, prevention, and improved management.

Key aspects include:

- **Waste Audit & Data Collection:** Campuses are encouraged to examine waste figures and conduct detailed audits to identify types of waste generated (food waste, recyclables, hazardous waste, electronic waste, etc.).
- **Waste Prevention:** Institutions should highlight measures already in place for waste reduction (e.g., reusable materials, digitalisation, composting) and consider opportunities for further prevention.

- Recycling & Diversion from Landfill: Evaluation of current recycling systems is required, along with identifying new streams (such as electronic waste, food waste, and lab materials) that can be diverted from landfill.
- Hazardous Waste Management: Correct procedures for the disposal of laboratory chemicals, paints, oils, batteries, and other hazardous materials must be verified and supported with proper documentation.
- E-Waste Compliance: Colleges must ensure that used electrical and electronic equipment is disposed of through proper channels in compliance with Waste Electrical and Electronic Equipment (WEEE) regulations.
- Litter Management: Campuses should assess litter hotspots, identify patterns of heavy littering (e.g., during festivals, rag weeks, open days), and implement preventive measures. Litter surveys and staff/grounds interviews are recommended to identify practical solutions.

Monitoring Indicators:

Common environmental performance indicators used in campuses include:

- Kilograms of waste per student (FTE) annually.
- Percentage of waste recycled.
- Kilograms of waste prevented through initiatives.

Composting Leaves into Nutrient-Rich Fertilizer

Composting fallen leaves is an efficient and environmentally friendly way to transform yard waste into valuable fertilizer that enriches soil health.

Rather than letting leaves go to landfill, they can be repurposed to create a natural, sustainable resource for campus greenery.

1. Preparing the Leaves

- Shredding accelerates decomposition—use a lawn mower or leaf shredder to break leaves into smaller pieces; if tools aren't available, layering whole leaves works too, though more slowly.
- Balance Browns and Greens—leaves are 'browns' (high in carbon). Mix them in a ratio of about 3 to 4 parts browns to 1 part greens (like grass clippings or kitchen scraps) to provide nitrogen and speed up composting.

2. Building the Compost Pile

- Layering method: Start with a coarse layer (twigs or shredded leaves for airflow), then alternate layers of browns and greens. Maintain moisture similar to a damp sponge.
- Pile Size: Aim for at least 3–4 ft wide and tall; too small, and it won't heat up; too large, and inner layers may become anaerobic.

3. Maintaining the Compost

- Aeration: Turn the pile 1–2 times per week. This introduces oxygen, boosting microbial activity and speeding decomposition.
- Microbial Heat: A healthy pile heats up internally. If it doesn't, check moisture, airflow, and balance.

- Covering the pile can help retain warmth and moisture, and prevent it from drying out, which is especially useful during hot seasons

4. Compost Readiness & Use

- Finished compost is dark, crumbly, and earthy-smelling. It's ready in a few weeks with active management, or several months if left to decompose passively.
- You can also let leaves decompose slowly into leaf mold, a fine, moisture-retaining soil conditioner ideal for mulching.

5. Benefits of Leaf-Based Compost

- Soil fertility and structure improve with nutrient-rich humus and microbial activity. Compost adds nitrogen, phosphorus, potassium, and beneficial microbes.
- It helps soils retain water and reduces dependence on synthetic fertilizers.
- Composting also reduces landfill waste and cuts carbon emissions.

Strategies and challenges for green campus:

1. Water Conservation and Rainwater Management

Sustainable campuses adopt water-saving techniques such as rainwater harvesting (RWH), efficient irrigation, and drought-resistant landscaping. For example, smart irrigation controllers that adjust watering based on weather and soil data significantly reduce water usage.

2. Institutional Framework and Stakeholder Engagement

A successful green campus requires a dedicated Sustainability Office, strong leadership, funding, and active engagement from students, faculty, and staff.

Sustainability initiatives are most effective when embedded into curricula and supported by broad participation across campus stakeholders.

Relevance to the DA-IICT Green Campus Management System

These insights directly inform system's design by highlighting important system requirements:

- Eco-gardening Module: The database should support tracking of eco-garden initiatives, offering educational and biodiversity metrics.
- Water Resource Management Integration: Add modules to log water-saving techniques, rainwater harvesting data, and irrigation schedules for informed decision-making and reporting.
- Governance and Engagement Tools: Incorporate features that facilitate oversight (Sustainability Office dashboards), multi-role access, and participation tools for staff and students.

Suitability of Rainwater Harvesting at DA-IICT:

Difficulties Faced Without This Process:

- Water shortage for irrigation during hot and dry months.
- Increased dependence on municipal/external water supply, raising costs.
- Unequal distribution of water across the plantation area, stressing certain zones.
- Maintenance burden on gardening staff due to inconsistent supply.

When We Will Approach This Process:

- During summer season (March–June), when water demand is highest and natural availability is lowest.
- When rainfall is below average or delayed, creating pressure on groundwater resources.
- During plantation drives or the addition of new flora, which increases irrigation needs.
- When groundwater levels decline due to overuse in Gandhinagar's soil and climatic conditions.

Why Rainwater Harvesting is the Solution:

- Provides a sustainable backup water source during peak summer demand.
- Helps in recharging groundwater and improving soil moisture levels.
- Reduces operational costs by minimizing external water dependency.
- Ensures long-term survival and health of campus flora even in extreme weather.
- Generates data (via the DBMS) on rainfall vs harvested water, useful for sustainability research.

2.1.2 References:

- [1] [DA-IICT, *Tree Survey at DA-IICT*. Gandhinagar: Dhirubhai Ambani Institute of Information and Communication Technology, 2019.](#)
- [2] [University of Windsor, *UWin Tree Management Plan \(v1.0\)*. Windsor: University of Windsor, 2020.](#)
- [3] [Article By Amitabh Teotia for DAIICT green campus design](#)
- [4] [Composting by leaves By Epic Gardening](#)
- [5] [Green Campus Programme By Green-Campus-Ireland](#)
- [6] [Strategies and challenges for green campus By Frontiers](#)
- [7] [Harvesting water from waterrainharvesing](#)
- [8] Water Harvesting Youtube video:
https://youtu.be/ICX6g4RuEfk?si=7_S7oBPAeNZ8vZBS
https://youtube.com/shorts/idR_BE0eM0Y?si=WmsCmT8ahYf3ndR0
- [9] [Google colab link for pie chart for trees types and number of percentage for that](#)

2.1.3 Combined Requirements from Background Readings

1. Eco-Gardening and Biodiversity Management

- Database support for tree inventory (species, indigenous varieties, locations, age, survival status).
- Tracking of herb gardens and medicinal plants with educational/research use data.
- Monitoring of bird species diversity and biodiversity indicators on campus.
- Recording landscape features (lawns, bamboo plantations, wild grasses, benches, fountains, lakes).

2. Water Conservation & Rainwater Harvesting

- Module to log and analyze rainwater harvesting data (collection, storage, usage).
- Integration of irrigation schedules with recycled water availability.
- Tools for monitoring water usage efficiency and cost savings vs. municipal supply.
- Data collection on seasonal water demand (e.g., summer, plantation drives).
- Reports on groundwater recharge and soil moisture levels.

3. Litter & Waste Management

- Waste Audit Module: Track amounts/types of waste (food, recyclables, hazardous, e-waste).
- Prevention Tools: Record initiatives such as composting, digitization, reusable materials.
- Recycling Data: Monitor waste diverted from landfill, percentage recycled.
- Hazardous Waste Handling: Compliance logging for chemicals, oils, batteries, lab waste..
- Litter Monitoring: seasonal spikes (festivals, open days), and mitigation actions.

4. Composting & Fertilizer Production

- Module to log leaf composting activities (collection, pile size, layering method).
- Monitoring of carbon/nitrogen balance (browns vs greens) in compost.
- Record compost readiness timeline and output quantities.
- Link compost data to soil health improvements (fertility, water retention).
- Option to track reduction in synthetic fertilizer dependency.

5. Sustainability Governance & Stakeholder Engagement

- Dashboard for Sustainability Office with consolidated indicators.
- Multi-role access (students, staff, faculty, gardening team, admin).
- Modules for student participation in eco-initiatives and reporting (projects, audits).
- Curriculum Integration: Linking green practices with academic and research modules.

2.2 Interviews

Green Campus Management: Interview Plan

System: DAIICT Green Campus Management System

Project Reference : GCM/DAU/2025/02

Participants: Vinas Moradiya, Preet Hirapara - Project Developers

Mr. Vishal (Green Campus Supervisor)

Date: 01/09/2025 **Time:** 17:00

Duration: 30 minutes **Place:** Vishal's Office

Purpose of Interview:

To gather detailed requirements and identify challenges for the proposed Green Campus Management System, focusing on:

- Tree Plantation Tracking (unique tree IDs, health data, location mapping)
- Composting & Fertilizer Management (from dry leaves & organic waste)
- Rainwater Harvesting Integration
- Revenue Generation Opportunities (wood/branches, flowers, organic compost)
- Optimizing campus greenery maintenance

Agenda:

- Current methods of managing plantation and tree data.
- Issues faced in tree identification, maintenance, and record-keeping.
- Requirements for a system that assigns **unique tree IDs** linked to detailed tree information pages.
- Expected reports (water usage, compost production, revenue reports).

Documents to be brought to the interview :

- Reports related to green campus management

Green Campus Management: Interview Plan

System: DAIICT Green Campus Management System

Project Reference : GCM/DAU/2025/02

Participants: Vinas Moradiya, Preet Hirapara - Project Developers

Mr.Harshad (Wormy Composting Site Supervisor)

Date: 01/09/2025 **Time:** 16:00

Duration: 40 minutes **Place:** Wormy Compost Site

Purpose of Interview:

To gather detailed requirements and identify challenges for the making of wormy compost and method for making it.

Agenda:

- Introduction & Orientation
- Process Demonstration
- Materials & Equipment Used
- Challenges & Solutions
- Revenue Opportunities
- Discussion & Q&A

Summary of interviews:

Interview with Mr.Vishal

In the interview with Mr. Vishal, we discussed the importance of maintaining a green campus and how it contributes to both environmental sustainability and the overall atmosphere of DAIICT. He explained the number of staff members involved in managing the greenery and the different equipment used for day-to-day maintenance.

He highlighted how tasks are distributed among the staff, such as watering, pruning, lawn trimming, and nursery management. We also explored the different methods of pruning, including *soft pruning* for routine maintenance and *hard pruning* for controlling plant growth and shaping trees.

Additionally, Mr. Vishal emphasized the importance of lawn trimming for keeping the campus neat and visually appealing. He shared insights about the various species of trees planted on campus and how the nursery supports propagation techniques like seed germination, grafting, and cutting for producing healthy new plants.

A very interesting point he mentioned was the role of trees in campus security. For example, species like *Bougainvillea* (boganvell), when planted along the boundary, form a natural protective barrier due to their thorny structure, reducing the chance of trespassing. This highlights how plantation also serves functional purposes beyond aesthetics.

Interview with Mr. Harshad

In the interview with Mr. Harshad, the focus was on the composting process and fertilizer production at DA-IICT. He explained the raw materials used, mainly dry leaves, which are first crushed before being processed. These crushed leaves are then mixed in a machine along with certain beneficial bacteria and controlled amounts of chemicals to accelerate decomposition.

The material is then arranged into piles and watered regularly for around 45 days, during which the decomposition takes place. After this period, the compost undergoes separation to remove any unwanted waste material. Finally, the prepared organic fertilizer is collected, packed, and even sold, generating an additional revenue stream for the campus.

Mr. Harshad also emphasized how this process not only reduces waste but also produces a high-quality organic fertilizer, which is eco-friendly, improves soil fertility, and supports the sustainability goals of the institute.

2.3 Questionnaires

Green Campus Management System

Please circle your answers to the following questions:

1. How would you rate the importance of green spaces in a university campus?

Extremely Important / Important / Neutral / Not Important

2. Do you think DA-IICT's greenery is currently well managed? Yes/No/Maybe

3. Should every tree have a unique digital ID for monitoring? Yes/No

4. Which information should be stored for each tree? (Multiple Choice)

Species name/Date of plantation/Location (GPS)/Growth stage/Height/Health Condition/Fertilizer and water requirements

5. How frequently should data (growth, health, etc.) of trees be updated?

Weekly/Monthly/Quarterly/Only when required

6. What resources are most important to track digitally? (Multiple Choice)

Fertilizers/Gardening tools and equipment/Staff schedules and duties/Water usage

7. Do you think that we can create compost from mess and cafeteria wet wastages ?

Yes/No/Maybe

8. Do you think that we can also sell flowers? Yes/No

9. Do you think that we also have to sell flower plants which produced by our campus' nursery? Yes/No/Maybe

10. How often do you do soft pruning? Daily/Weekly/Monthly

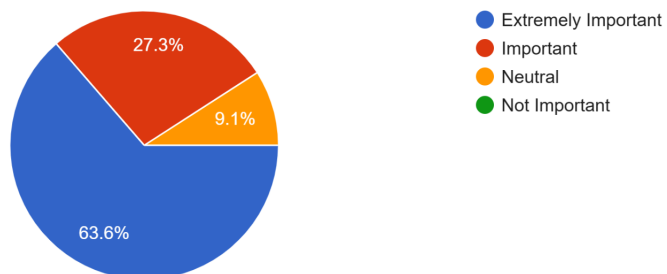
11. How often do you do loan trimming? Weekly/Biweekly/Monthly

Give some suggestion about maintaining green campus? _____

[Google Form Link](#)

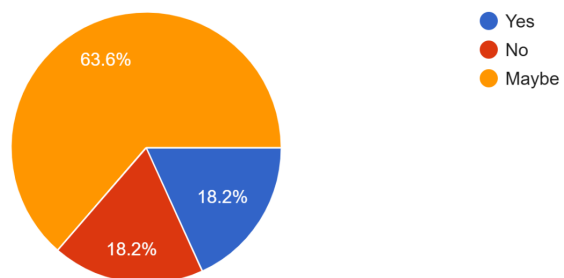
How would you rate the importance of green spaces in a university campus?

11 responses



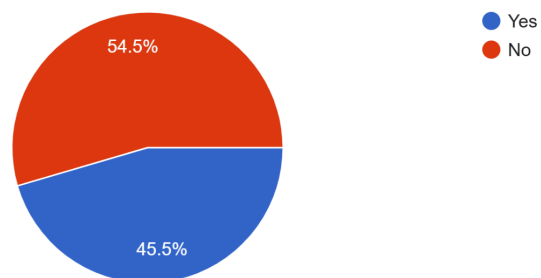
Do you think DA-IICT's greenery is currently well managed?

11 responses



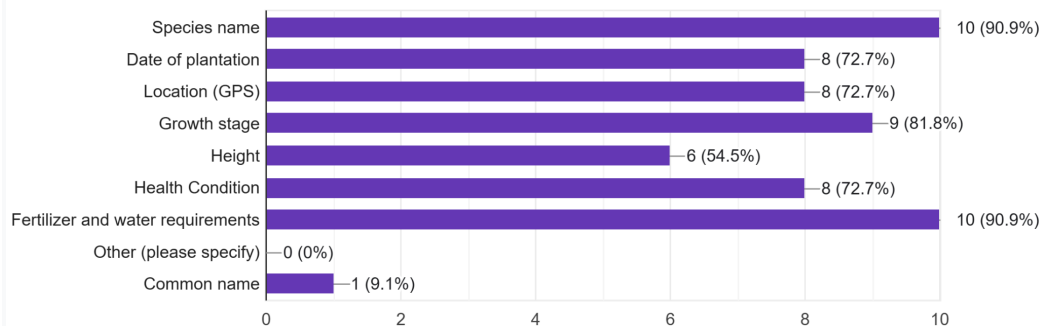
Should every tree have a unique digital ID for monitoring?

11 responses



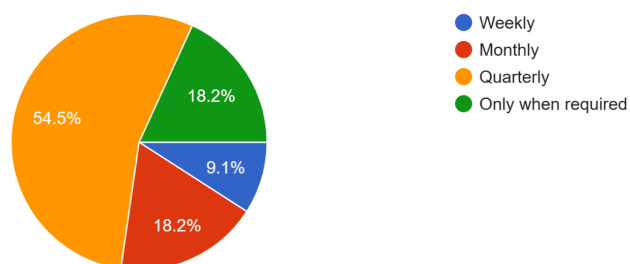
Which information should be stored for each tree? (Multiple Choice)

11 responses



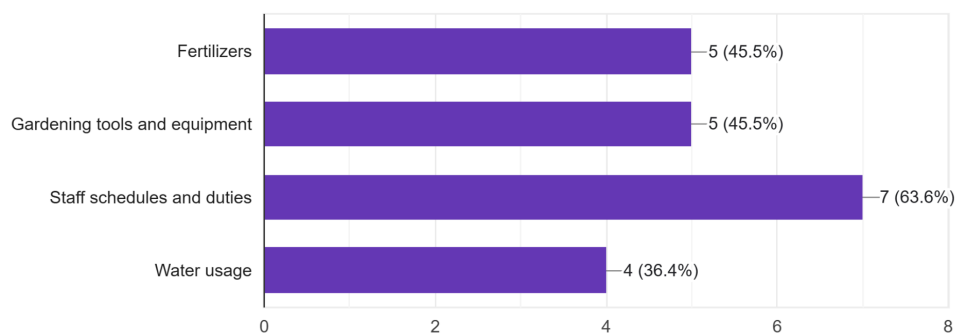
How frequently should data (growth, health, etc.) of trees be updated?

11 responses



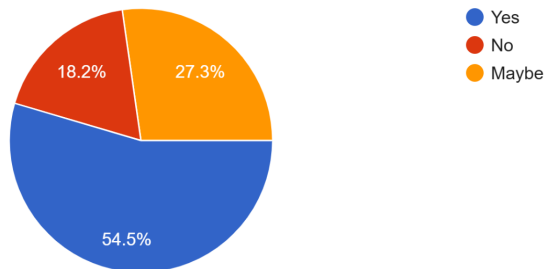
What resources are most important to track digitally? (Multiple Choice)

11 responses



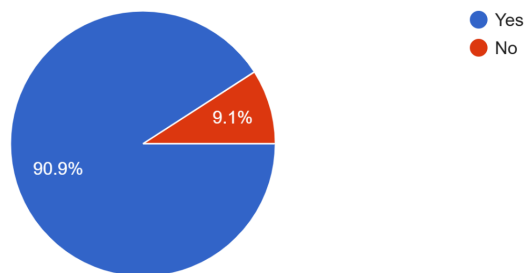
Do you think that we can create compost from mess and cafeteria wet wastages ?

11 responses



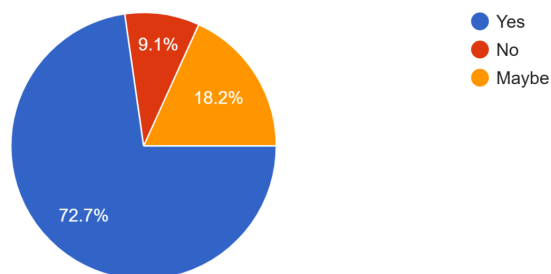
Do you think that we can also sell flowers?

11 responses



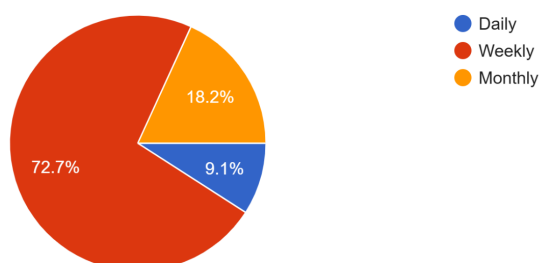
Do you think that we also have to sell flower plants which produced by our campus' nursery?

11 responses



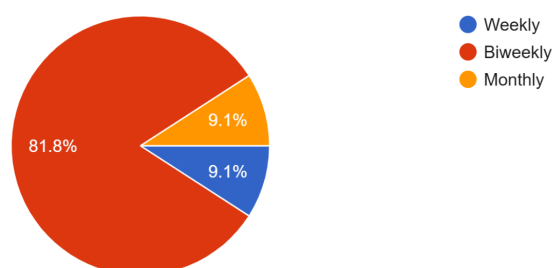
How often do you do soft pruning?

11 responses



How often do you do loan trimming?

11 responses



Requirements:

- Maintain and enhance green spaces as a core part of DA-IICT's identity.
- Improve and optimize current greenery management practices to reach consistent satisfaction among stakeholders.
- Implement a tree identification system (digital IDs with QR codes or tags) for better monitoring and tracking.
- Create a Tree Database System with these standardized attributes.
- Develop a Resource Tracking System integrated with tree management.
- Implement a composting system to convert wet waste into organic fertilizer for campus use.
- Explore revenue generation models by selling flowers, saplings, and nursery plants produced on campus.
- Standardize maintenance schedules for hard pruning and soft pruning
- Create a comprehensive Green Campus Management Plan combining monitoring, composting, and revenue opportunities.

3.Fact Finding Chart

Objective	Technique	Subject	Time Commitment
To understand the importance of a green campus and its role in DA-IICT's identity	Interview	Mr. Vishal	15 minutes
To gather details about staff numbers, roles, and equipment used for maintenance	Interview	Mr. Vishal	10 minutes
To study the role of different species (e.g., bougainvillea as security around borders)	Interview/ Observation	Mr. Vishal	20 minutes
To understand composting methods used on campus	Interview	Mr. Harshad	20 minutes
To gather information on raw materials for composting (e.g., dry leaves, bacteria, chemicals)	Interview	Mr. Harshad	5 minutes
To learn about the composting process (crushing, mixing, piling, watering for 45 days, waste removal, fertilizer packing & selling)	Interview	Mr. Harshad	12 minutes
To establish requirements for digital monitoring of trees and greenery (IDs, health, fertilizer, growth data)	Questionnaire	Staff/ Student	1 hour
To explore revenue opportunities from flowers, nursery plants, and compost fertilizer	Interview/ Questionnaire	Nursery staff, Students	1.5 hour

4. List Requirements

- The system should assign a unique digital ID to every tree on campus.
- The system should store tree details such as species name, common name, date of plantation, GPS location, growth stage, height, health condition, and fertilizer/water requirements.
- The system should allow regular updates (weekly, monthly, quarterly, or yearly) of tree data such as growth, health, and maintenance activities.
- The system should track staff duties and schedules related to gardening, pruning, trimming, and composting.
- The system should log resources used such as fertilizers, gardening tools, equipment, and water usage.
- The system should include modules for pruning management (hard pruning, soft pruning, and lawn trimming with their frequency).
- The system should support rainwater harvesting tracking, especially useful in summer when water usage is high.
- The system should maintain data about composting processes, including input materials (dry leaves, wet waste), machines used, bacteria/chemicals added, and duration (e.g., 45 days).
- The system should generate reports on compost produced, fertilizer usage, and waste reduced.
- The system should include functionality to track nursery activities such as propagation of plants, seedling growth, and availability of saplings.
- The system should track and support flower and plant selling activities, including types of flowers grown, sales, and revenue generated.
- The system should store and manage security contributions of plants (e.g., bougainvillea on campus borders).

- The system should allow observation data entry from field staff (manual entry after field visits).
- The system should support integration with digital platforms (like QR codes) for quick access to tree/plant information.
- The system should provide analysis and visualization tools (graphs, charts) for monitoring campus greenery trends.

5. User Classes and Characteristics

1. Admin

- Full rights over the system.
- Manages all records, user accounts, and generates reports for higher authorities.

2. Gardening Staff

- Enter observation data (watering, pruning, tree health, composting).
- Update usage of equipment and fertilizers.

3. Supervisor

- Monitors gardening staff activities.
- Validates field data and assigns new tasks.

4. Nursery Manager

- Maintains records of saplings, flower plants, and propagation details.
- Tracks availability of plants for plantation or sale.

5. Sales Manager

- Manages stock of compost, flowers, and fertilizers.
- Records sales and prepares simple revenue reports.

6. Security Staff

- Checks plantation along campus boundaries.
- Reports issues like damaged fencing or missing plants (view/update limited).

7. Students / General Users / Faculty

- View tree information through digital IDs/QR codes.
- Contribute feedback or suggestions for campus greenery (read-only access).

6. Operating Environment

6.1 Hardware/Software Requirements

- Hardware : Any standard PC for admin, Smart phones for staff
- Software : PostgreSQL
- OS : Windows/Linux/Mac

6.2 External Interfaces

- Input : SQL queries / forms
- Output: Query results, reports
- Possible third-party: Google Forms (for questionnaire)

7. Product Functions

CRUD operations:

- Trees (species, plantation date, health, fertilizer usage)
- Staff (roles, tasks, schedules)
- Equipment (type, usage logs, maintenance)
- Compost/Fertilizer batches (status, quantity, usage)
- Nursery plants (stock, propagation, sales)
- Sales (flowers, compost, saplings)

Triggers:

- Auto-update tree health status if new observation is entered
- Auto-reduce inventory stock when flowers/fertilizers are sold
- Notify Supervisor if a tree's health is marked "critical"
- Update compost batch status when 45 days are completed

Views:

- View all healthy/critical trees
- View nursery stock ready for sale
- View weekly staff schedules and completed tasks
- View total revenue generated from sales

Stored Procedures:

- Calculate fertilizer/water requirements per tree based on species
- Generate monthly reports of tree growth and staff activities
- Generate compost production summary (batches completed vs in-process)
- Calculate total revenue from flower, sapling, and compost sales

8. Privileges

Function	Admin	Gardening Staff	Supervisor	Nursery Manager	Sales Manager	Security Staff	Students / Users
Add/Delete Users	Yes	No	Yes	No	No	No	No
Manage Tree Records (CRUD)	Yes	Yes (limited)	Yes	No	No	View only	View only
Manage Staff Records	Yes	No	Yes	No	No	No	No
Enter Observation Data	Yes	Yes	Yes	No	No	Limited	No
Manage Equipment Records	Yes	Usage only	Yes	No	No	No	No
Manage Compost/Fertilizer Records	Yes	Entry only	Yes	No	Yes	No	No

Manage Nursery Stock	Yes	No	Yes	Yes	No	No	No
Manage Sales Records	Yes	No	No	No	Yes	No	No
Generate Reports (growth, revenue)	Yes	No	Yes	Yes	Yes	No	No
View Tree Info (Digital ID/QR)	Yes	Yes	Yes	Yes	Yes	Yes	Yes

9. Assumptions

- Users (Admin, Staff, Supervisors) have basic computer and mobile skills to operate the system.
- Internet connection or campus Wi-Fi will be available for syncing data to the central database.
- Field staff will have access to mobile devices for data entry.
- The dataset will be moderate in size (few thousand tree/plant records, manageable for academic use).
- The system will be initially deployed for DA-IICT campus only, with scope for expansion later.
- Data collected from staff and surveys will be reliable and accurate.
- Hardware (PCs, smartphones, servers) will already be available within the institute.

10. Business Constraints

- Time: Must be developed within the semester timeline.
- Cost: Use free DBMS (e.g., PostgreSQL).
- Skill: Students' knowledge limited to DBMS course.
- Scope: The project will be implemented as a prototype focusing mainly on DA-IICT campus, not as a large-scale commercial system.

Summary of SRS for DAIICT Green Campus Management System:

(1) Requirements/Why we need to:

Before doing any work or project, we should think about what kind of results this project Will give.

Is it worth doing?

Is it helpful for audience?

Will it improve decision-making for administration or audience?

In our project, the answer to all these question is Yes.

The next step is to know which types of people will use this management system and how it will be helpful for them compared to the current system.

- Category Of People (Intended Audience)
- Campus Management Staff
- DAIICT Administration
- Researchers and Students
- Database Developers / Project Team
- Nursery Manager
- Sales Manager

(2) Primary Objectives are:

- Systematic management of plantation data (trees, shrubs, herbs).
- Efficient tracking of fertilizer and equipment usage.
- Recording staff roles, duties, and schedules.
- Supporting eco-friendly initiatives like composting of leaves and cafeteria waste.
- Also we can improve our campus security by tracking shrubs, vines details at the boundaries of campus.
- Also make revenue generation through sale of flowers, fertilizers, and wood by-products.
- Add QR codes so people can easily scan and see information about each tree.

(3) Key features of this Management system:

- Every tree gets a unique digital ID to help track it easily.
- The system stores important details about each tree, like its type, botanical name, common name, GPS location, health, growth stage, and fertilizer used.
- It helps to manage gardening staff schedule, equipment and work progress.
- Also there is a composting feature that converts dry leaves and organic waste into natural fertilizer.
- System can manage selling flowers, grafted plants which are made from DAIICT nursery.
- Rainwater harvesting tracking, It tracks rainwater collection, which is helpful in summer when there is less water available.
- It gives useful reports about tree health, resource use, sales, and how eco-friendly the system is.

document uses ideas from articles about green campus design and composting, talks with green campus management supervisors (Mr. Vishal) and gardening staff member (Mr. Harshad) and surveys from staff. These helped us figure out the most important features the system should have

- Track plant health, growth, and maintenance digitally.
- Manage fertilizers and gardening tools in an organized way.
- Help make smart decisions to support sustainability.
- Contribution of shrubs and vines in our campus security.
- Staff use the system easily.

The system has clear user roles, giving specific access to Admin, Gardening Staff, Supervisor, Nursery Manager, Sales Manager, Security Staff, and General Users. It's built in separate parts (modules) so it's easy to update or expand later.

It's made to be low-cost by using free software (like PostgreSQL), work on normal computers, and fit the project limits of a semester assignment. The system is a prototype for DAIICT campus but can be expanded in the future.

Overall, this system helps DAIICT run a greener campus, manage work better, and support eco-friendly practices.